GRADE 10 / Mathematics / English

Factorise, solve, check

01 Understand

Learning goals: Extract a common factor • Factorise a quadratic trinomial • Verify roots by substitution

Factorising reverses multiplication. Always check for a common factor before trying brackets.

For $x^2 + bx + c$, find two numbers with sum b and product c. This shortcut assumes the coefficient of x^2 is 1.

The zero-product rule applies to a product equal to zero: AB = 0 implies A = 0 or B = 0.

Build the brackets

Factor pair	Sum	Product
1 and 6	7	6
2 and 3	5	6
−2 and −3	−5	6

02 Follow the reasoning

Solve $x^2 - 5x + 6 = 0$.

1. The numbers −2 and −3 have sum −5 and product 6.

2. Write $(x - 2)(x - 3) = 0$. Set each bracket equal to zero.

3. $x = 2$ or $x = 3$. Check: $4 - 10 + 6 = 0$ and $9 - 15 + 6 = 0$.

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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. Factorise $6x^2 + 9x$.

2. Solve $x^2 + 7x + 12 = 0$.

3. A learner says $x^2 = 9$ gives only $x = 3$. Correct the answer and explain.

04 Check and correct

1. $3x(2x + 3)$. The common factor is $3x$.

2. $(x + 3)(x + 4) = 0$, so $x = -3$ or -4.

3. $x = 3$ or -3, because squaring either number gives 9.

Common mistake to avoid

Do not cancel terms across addition. Only common factors can be cancelled in a fraction.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Algebraic expressions; equations and inequalities

DBE CAPS, Grades 10–12: Mathematics. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID math-10.